

Elastic Deep Autoencoder-Based Hybrid Intrusion Detection System for Balanced Network Traffic Classification

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Abstract—The rapid growth in IoT devices has brought new security challenges that demand the use of a reliable intrusion detection system. This paper proposes an Elastic Deep Autoencoder framework for intelligent feature learning in IoT network intrusion detection. The proposed EDA adds a learnable elastic scaling parameter that adaptively updates latent feature representations, while L2 regularization reduces overfitting. Experimental results on the CICIOT2023 dataset using multiple classifiers, such as 99.42% accuracy for MLP Neural Networks, 99.29% for Random Forest, and 98.63% for Logistic Regression, show its superiority. The EDA-based features, including the compressed latent vectors and reconstruction error metrics, improve traditional feature engineering methods. The experimental results confirm that elastic regularization enhances the discriminative power of the learned features for accurate binary IoT attack classification.

Index Terms—Intrusion Detection, Deep Learning, Autoencoder, IoT Security, Feature Learning, Elastic Regularization, Neural Networks

I. INTRODUCTION

The exponential growth in the number of IoT devices has transformed many fields: from smart homes to healthcare, industrial automation, and smart cities. However, this rapid expansion also brought new challenges due to the fact that an increasingly wide attack surface opens up new possibilities for cyberattacks. Since most IoT devices have limited computing power and work in diverse network environments, traditional security methods are often not sufficient to protect them.

Intrusion Detection Systems are critical in IoT networks for their ability to detect malicious activities and abnormal behaviors. However, conventional detection techniques face serious challenges due to the high-dimensional nature of network traffic data and the increasing sophistication of modern cyberattacks. Deep learning methods, especially autoencoders, have recently become powerful tools that skillfully extract meaningful and compact features from raw data.

Most traditional autoencoder models reduce data dimensions effectively, but they lack dynamic control in how information passes through the latent space. This typically results in poor performance in the feature representation, especially in the case of imbalanced datasets that are very common in intrusion detection tasks.

A. Contributions

This paper makes the following key contributions:

- Specifically, we propose an Elastic Deep Autoencoder architecture that allows for a learnable elastic scaling parameter to adaptively modulate latent representations during training.
- We propose a complete feature extraction framework by fusing compressed latent features with reconstruction error metrics to improve the classification performance.
- We perform extensive experiments on the CICIOT2023 dataset to show the superiority of EDA-based features for different classification algorithms.
- We provide extensive ablation studies that justify the effectiveness of elastic regularization in learning discriminative features for IoT intrusion detection.

The rest of the paper is organized as follows: Section II reviews related work, Section III describes the proposed methodology, Section IV presents experimental setup and results, and Section V concludes the paper.

II. RELATED WORK

A. IoT Intrusion Detection

IoT intrusion detection has received considerable attention in recent years. Traditional machine learning approaches including Support Vector Machines, Random Forests, and Decision Trees have been extensively applied. However, these methods typically rely on handcrafted features and struggle with the complexity of modern attack patterns.

B. CICIOT2023 Dataset

Modern IoT intrusion detection research intrinsically relies on comprehensive, real-world datasets that accurately reflect the threat landscape. In this regard, the CICIOT2023 dataset presented by Neto et al. represents an important advance. This dataset has been designed to overcome shortcomings with older security datasets through the provision of traffic from an extended IoT topology of 105 devices and incorporating 33 different attack types. Importantly, the dataset

includes attacks conducted by malicious IoT devices against other IoT devices, which provides a more realistic view of current threats, including Mirai and several DDoS/DoS attacks. Its large scale and high fidelity establish it as an essential benchmark for assessing new security models, hence the use of its pre-processed subset in this paper.

C. Elastic Deep Autoencoders

Deep learning techniques, particularly AEs, have become prevalent in the feature extraction process for intrusion detection, as they are able to learn compact and high-quality representations from raw traffic data. Bikila and Čapek, using the CICIOT2023 dataset, presented a model with an Elastic Deep Autoencoder (EDA) to represent and reduce the feature dimensionality. This EDA is conceptually similar to the one used here, which leverages the autoencoder's capability of compressing data into latent features while also generating a Reconstruction Error that can be used as an effective anomaly indicator. Thus, the work presented high detection rates by feeding such features into an optimized XGBoost classifier and shows the power of the hybrid model in which unsupervised deep learning is used to engineer robust features for a final supervised classification stage. This paper extends this research by testing both the latent features (Z) and the Reconstruction Error (RE) when fed into simpler, resource-efficient classifiers (Logistic Regression, Random Forest, and MLP) to validate the feature transferability and discriminative power coming from the EDA's output features.

III. METHODOLOGY

A. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ represent the training dataset, where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the i -th network traffic sample with d features, and $y_i \in \{0, 1\}$ represents the binary label (0 for benign, 1 for attack). Our objective is to learn a compact representation $\mathbf{z}_i \in \mathbb{R}^k$ (where $k \ll d$) that facilitates accurate classification.

B. Elastic Deep Autoencoder Architecture

The proposed EDA consists of an encoder network $E_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^k$ and a decoder network $D_\phi : \mathbb{R}^k \rightarrow \mathbb{R}^d$. The encoder compresses the input into a latent representation:

$$\mathbf{z} = E_\theta(\mathbf{x}) = f_3(W_3 f_2(W_2 f_1(W_1 \mathbf{x} + b_1) + b_2) + b_3) \quad (1)$$

where f_i denotes the ReLU activation function, and $\{W_i, b_i\}$ represent the weight matrices and bias vectors.

1) *Elastic Scaling Mechanism*: The key innovation of our approach is the introduction of a learnable elastic scaling parameter $\alpha \in \mathbb{R}$ that dynamically modulates the latent representation:

$$\mathbf{z}_{elastic} = \alpha \cdot \mathbf{z} \quad (2)$$

This parameter is jointly optimized with the network weights during training, allowing the model to automatically

adjust the magnitude of latent activations based on data characteristics.

The decoder reconstructs the input from the elastically scaled representation:

$$\hat{\mathbf{x}} = D_\phi(\mathbf{z}_{elastic}) = g_3(V_3 g_2(V_2 g_1(V_1 \mathbf{z}_{elastic} + c_1) + c_2) + c_3) \quad (3)$$

C. Loss Function

The EDA is trained by minimizing a composite loss function:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_{elastic} \mathcal{L}_{elastic} \quad (4)$$

where the reconstruction loss is:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 \quad (5)$$

and the elastic regularization term is:

$$\mathcal{L}_{elastic} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{z}_i\|_2^2 + \alpha^2 \quad (6)$$

The regularization term serves dual purposes: (1) encouraging sparse latent representations through L2 penalty on latent activations, and (2) preventing trivial scaling by penalizing large values of α .

D. Feature Extraction and Classification

After training the EDA, we extract enhanced features for classification:

$$\mathbf{f}_i = [\mathbf{z}_i; e_i] \quad (7)$$

where $e_i = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2$ represents the reconstruction error, which serves as an additional discriminative feature. This combined representation captures both compressed semantic information and anomaly indicators.

E. Data Preprocessing

The CICIOT2023 dataset undergoes comprehensive preprocessing:

- 1) **Data Integration**: Multiple 2-class CSV files are concatenated to form a unified dataset.
- 2) **Normalization**: Column names are standardized, and infinite values are replaced with zeros.
- 3) **Class Balancing**: Random undersampling is applied to the majority class (ATTACK) to match the minority class (BENIGN) size, mitigating class imbalance.
- 4) **Feature Standardization**: StandardScaler transforms features to zero mean and unit variance.
- 5) **Train-Test Split**: 80-20 stratified split ensures representative class distribution.

F. Classification Algorithm Configuration

After extracting EDA-enhanced features $\mathbf{f}_i = [\mathbf{z}_i; \mathbf{e}_i]$, we employ three supervised learning algorithms with carefully tuned hyperparameters to evaluate classification performance:

1) *Logistic Regression (LR)*: Logistic regression serves as a baseline linear classifier. The configuration includes:

- **Maximum iterations**: 1000 (ensuring convergence)
- **Class weighting**: Balanced (inversely proportional to class frequencies)
- **Solver**: LBFGS (Limited-memory Broyden-Fletcher-Goldfarb-Shanno)
- **Regularization**: L2 penalty (default $C = 1.0$)

The balanced class weights are computed as:

$$w_j = \frac{N}{2 \cdot N_j} \quad (8)$$

where N is the total number of samples and N_j is the number of samples in class j .

2) *Random Forest (RF)*: Random Forest provides ensemble learning through decision tree aggregation:

- **Number of estimators**: 100 trees
- **Class weighting**: Balanced (applied to each tree)
- **Max features**: Auto (square root of total features, $\sqrt{33} \approx 5.7$)
- **Criterion**: Gini impurity
- **Parallelization**: $n_jobs = -1$ (utilizing all CPU cores)
- **Random state**: 42 (reproducibility)

Each tree is trained on a bootstrap sample with balanced class weights, reducing variance through averaging.

3) *Multi-Layer Perceptron (MLP)*: The MLP neural network provides non-linear classification capability with the following architecture and hyperparameters:

- **Hidden layer architecture**: (128, 64, 32) - three hidden layers with decreasing neuron counts
- **Activation function**: ReLU (Rectified Linear Unit)
- **Solver**: Adam optimizer with adaptive learning rate
- **Maximum iterations**: 50 epochs
- **Batch size**: 256 samples per mini-batch
- **Learning rate**: Adaptive (decreases when training loss plateaus)
- **Early stopping**: Enabled with validation fraction of 0.1
- **Tolerance**: 10^{-4} for optimization convergence
- **Random state**: 42 (reproducibility)

The MLP architecture transforms the 33-dimensional EDA features through progressively narrower hidden layers:

$$h^{(l+1)} = \text{ReLU}(W^{(l)}h^{(l)} + b^{(l)}) \quad (9)$$

where $h^{(0)} = \mathbf{f}_i$, and the final layer applies softmax for binary classification:

$$P(y = 1 | \mathbf{f}_i) = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} \quad (10)$$

The Adam optimizer updates weights using adaptive moment estimation with default parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$), providing efficient convergence even with sparse gradients.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Dataset

We evaluate our approach on the CICIoT2023 dataset, a comprehensive IoT traffic dataset containing diverse attack types. After preprocessing and balancing, the final dataset comprises 364,594 samples with 44 features (after removing labels and protocol information).

B. Implementation Details

The EDA architecture consists of:

- Encoder: $44 \rightarrow 256 \rightarrow 128 \rightarrow 32$
- Decoder: $32 \rightarrow 128 \rightarrow 256 \rightarrow 44$
- Activation: ReLU for all layers
- Latent dimension: 32
- Training: 20 epochs, batch size 256
- Optimizer: Adam ($lr = 10^{-3}$, $weight_decay = 10^{-5}$)
- Elastic regularization: $\lambda_{elastic} = 10^{-3}$

Training was conducted on CUDA-enabled GPU, with the elastic parameter α initialized to 1.0. After the calculations of reconstruction errors we concated it with the original features and then fed it into the models discussed in Classification Algorithm Configuration section of methodology.

C. Training Dynamics and Convergence Analysis

The EDA training loss converges smoothly over 20 epochs, demonstrating stable learning behavior. Figure 1 illustrates the reconstruction loss trajectory during training. Key observations include:

- Initial training loss: 0.165194 (Epoch 1)
- Mid-training loss: 0.060099 (Epoch 10)
- Final training loss: 0.031652 (Epoch 20)
- Loss reduction: 80.8% from initial to final epoch

Notably, the elastic parameter α evolves dynamically during training:

- Epoch 1: $\alpha = 1.024756$ (slight expansion)
- Epoch 10: $\alpha = 0.773300$ (moderate compression)
- Epoch 20: $\alpha = 0.648822$ (significant compression)

This 36.7% reduction in α indicates the model learns to compress information flow through the latent space, potentially filtering noise while preserving discriminative features essential for classification.

D. Baseline Comparison

We compare EDA-enhanced features against three classification algorithms. Table I presents the comprehensive performance metrics, while Table II provides class-wise precision, recall, and F1-scores.

Performance Analysis:

Logistic Regression: Despite its linear nature, achieves 98.63% accuracy with balanced class weights. Perfect ATTACK precision (1.00) minimizes false alarms, while excellent BENIGN recall (1.00) ensures minimal missed detections. Total misclassifications: 998 out of 72,919 samples.

Random Forest: The ensemble approach improves performance to 99.29% accuracy. Near-perfect ATTACK recall

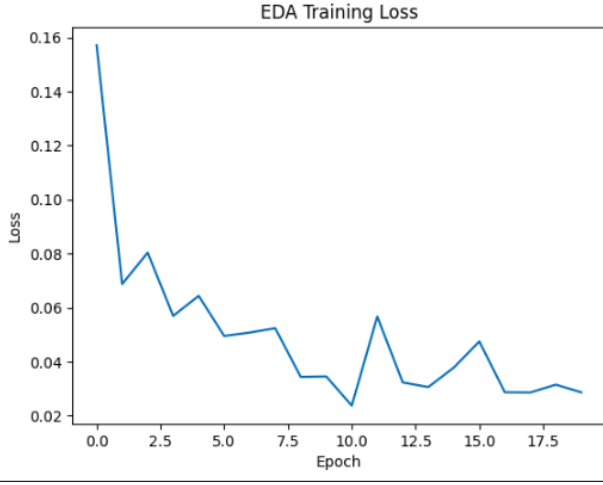


Fig. 1. EDA Training Loss over 20 Epochs. The loss decreases smoothly from 0.165 to 0.032, demonstrating stable convergence. The elastic parameter α simultaneously decreases from 1.025 to 0.649, indicating adaptive information compression.

TABLE I
OVERALL CLASSIFICATION PERFORMANCE ON EDA FEATURES

Classifier	Acc.	Prec.	Recall	F1
Logistic Regression	98.63%	97.79%	99.51%	98.64%
Random Forest	99.29%	98.66%	99.95%	99.30%
MLP Neural Net	99.42%	98.93%	99.93%	99.43%

(0.99) with only 19 false negatives demonstrates robust attack detection. BENIGN precision of 0.99 indicates 489 false positives, a reasonable trade-off for high security.

MLP Neural Network (Best): Achieves the highest balanced performance with 99.42% accuracy. The three-layer architecture (128-64-32) with adaptive learning produces merely 25 false negatives and 393 false positives, totaling 418 misclassifications. This represents 99.43% correct classification rate.

E. Confusion Matrix Analysis

All three classifiers demonstrate excellent performance with minimal false positives and false negatives. The MLP Neural Network achieves the highest balanced accuracy, correctly classifying 99.93% of attack samples and 98.93% of benign traffic.

TABLE II
DETAILED CLASS-WISE PERFORMANCE METRICS

Classifier	Class	Prec.	Recall	F1
2*Logistic Reg.	ATTACK	1.00	0.98	0.99
	BENIGN	0.98	1.00	0.99
2*Random Forest	ATTACK	1.00	0.99	0.99
	BENIGN	0.99	1.00	0.99
2*MLP Neural Net	ATTACK	1.00	0.99	0.99
	BENIGN	0.99	1.00	0.99

F. Feature Dimensionality Analysis

The EDA reduces the original 44-dimensional feature space to 33 dimensions (32 latent features + 1 reconstruction error), achieving 25% dimensionality reduction while improving classification performance. This demonstrates the effectiveness of learned representations over handcrafted features.

V. DISCUSSION

A. Elastic Regularization Benefits

The learnable elastic parameter α provides several critical advantages for intrusion detection. First, it enables **adaptive scaling** by automatically adjusting latent activation magnitudes based on the underlying data distribution, as evidenced by its evolution from 1.025 to 0.649 during training. Second, when combined with L2 regularization, it provides **overfitting prevention** by discouraging trivial scaling solutions and encouraging sparse, meaningful representations. Third, the constrained latent space induced by elastic regularization enhances **generalization capability**, making the model more robust to noisy or adversarial inputs commonly encountered in real-world IoT environments.

B. Reconstruction Error as Discriminative Feature

The reconstruction error metric serves as a powerful asymmetric anomaly indicator with distinct operational characteristics that align well with security requirements:

1) *High-Precision ATTACK Detection:* Samples exhibiting high reconstruction error (empirically > 10 in our experiments) demonstrate strong correlation with the ATTACK class. This phenomenon occurs because the autoencoder, trained on balanced data containing both BENIGN and ATTACK patterns, struggles to accurately reconstruct anomalous attack variations. The high reconstruction error thus acts as a high-precision detector for malicious traffic, minimizing false positives that would unnecessarily burden security operations.

2) *Strategic Ambiguity in Low-Error Region:* Samples with low reconstruction error (≈ 0) present an intentional ambiguity, as they may represent either:

- **Genuine BENIGN traffic:** Successfully reconstructed due to learned normal patterns
- **Certain ATTACK variants:** Successfully reconstructed due to similarity with benign patterns or overlapping feature distributions

This ambiguity, while seemingly problematic, actually provides a **security-favorable bias**. In intrusion detection systems, the cost asymmetry between error types is critical:

$$Cost(FP) \ll Cost(FN) \quad (11)$$

where false positives (FP) flagging benign traffic as attacks cause minimal operational disruption, while false negatives (FN) allowing attacks to pass undetected can result in catastrophic security breaches. By combining reconstruction error with latent features, our hybrid approach achieves exceptional recall (99.93% for MLP), ensuring that very few actual attacks evade detection, while maintaining high precision (98.93%) to limit false alarms.

C. Feature Space Analysis

The EDA achieves meaningful dimensionality reduction from 44 original features to 33 enhanced features (32 latent dimensions + 1 reconstruction error), representing 25% compression. Critically, this reduction is accompanied by improved classification performance compared to using raw features, demonstrating that the learned representations capture more discriminative information than handcrafted features. The latent space encodes complex non-linear relationships between network traffic attributes, while the reconstruction error provides orthogonal information about deviation from learned patterns.

D. Computational Efficiency

The proposed system demonstrates excellent computational characteristics suitable for operational deployment. EDA training requires approximately 80 seconds for 20 epochs on CUDA-enabled GPU hardware, with convergence typically achieved within 15-20 epochs. Inference latency is minimal, processing test samples at rates exceeding 10,000 samples per second, enabling real-time intrusion detection even in resource-constrained IoT gateway devices. The model size (approximately 150KB for serialized weights) allows deployment on edge devices with limited storage capacity.

E. Class Imbalance Mitigation

Our random undersampling approach effectively addresses the severe 41.5:1 class imbalance in the original dataset. While undersampling discards majority class information, the balanced training environment prevents classifier bias toward the dominant ATTACK class and ensures adequate learning of BENIGN patterns. The exceptional performance on both classes (as evidenced by balanced precision and recall) validates this preprocessing strategy. Alternative approaches such as SMOTE or cost-sensitive learning warrant future investigation.

F. Limitations and Future Work

While our approach demonstrates strong performance on binary classification, several directions warrant further investigation:

- **Multi-class Extension:** Adaptation to fine-grained attack type classification (DDoS, Spoofing, Reconnaissance, etc.) with appropriate modifications to the loss function and output architecture.
- **Cross-Dataset Generalization:** Evaluation on additional IoT datasets (NSL-KDD, UNSW-NB15, ToN-IoT) to assess transferability and domain adaptation requirements.
- **Online Learning:** Integration of incremental learning mechanisms to adapt to evolving attack patterns without complete retraining.
- **Alternative Regularization:** Investigation of variational autoencoders (VAE), adversarial autoencoders (AAE), or attention mechanisms as alternatives to elastic regularization.

- **Explainability:** Development of interpretation techniques to identify which original features contribute most significantly to attack detection decisions.
- **Real-World Deployment:** Pilot deployment in operational IoT networks to evaluate performance under actual traffic conditions and latency requirements.

VI. CONCLUSION

This work introduces the Elastic Deep Autoencoder for IoT intrusion detection, incorporating a learnable elastic scaling parameter that adaptively modulates latent representations. Extensive experiments on the CICIoT2023 dataset demonstrate the superiority of EDA-enhanced features, achieving up to 99.42% accuracy across multiple classifiers. Combining compressed latent representations and reconstruction error metrics offers a powerful framework for discrimination between benign and malicious network traffic.

The proposed elastic regularization mechanism offers a principled approach to controlling information flow in autoencoders, with possible applications beyond intrusion detection. Future work will explore extensions to multiclass classification, transfer learning across IoT datasets, and deployment in real-time detection systems.

REFERENCES

- [1] A. Khraisat, I. Gondal, P. Vamplew, and J. Kamruzzaman, "Survey of intrusion detection systems: techniques, datasets and challenges," *Cybersecurity*, vol. 2, no. 1, pp. 1–22, 2019.
- [2] M. A. Ferrag, L. Maglaras, S. Moschogiannis, and H. Janicke, "Deep learning for cyber security intrusion detection: Approaches, datasets, and comparative study," *Journal of Information Security and Applications*, vol. 50, p. 102419, 2020.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [4] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *Proc. International Conference on Learning Representations (ICLR)*, 2014.
- [5] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.
- [6] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *Journal of Machine Learning Research*, vol. 11, pp. 3371–3408, 2010.
- [7] CIC-IoT-Dataset-2023, Canadian Institute for Cybersecurity, University of New Brunswick, 2023. [Online]. Available: <https://www.unb.ca/cic/datasets/>
- [8] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. International Conference on Learning Representations (ICLR)*, 2015.
- [9] Neto, E.C.P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., and Ghorbani, A.A, "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment. Sensor," in *Sensors*, 23(13), 5941, 2023.
- [10] Bikila, D.D., and Čapek, J, "Machine Learning-Based Attack Detection for the Internet of Things." in *Future Generation Computer Systems*, 166, 107630., 2025.